Universidad Nacional del Litoral
Facultad de Ingeniería y Ciencias Hídricas

Examen Parcial 12/05/2023

1. Si $\nabla \cdot \mathbf{v} = 0$, $\nabla \times \mathbf{v} = \dot{\mathbf{w}}$ y $\nabla \times \mathbf{w} = -\dot{\mathbf{v}}$, donde $\dot{\mathbf{v}} = \dfrac{d\mathbf{v}}{dt}$ y $\dot{\mathbf{w}} = \dfrac{d\mathbf{w}}{dt}$, demostrar usando notación indicial que

$$\nabla^2 \mathbf{v} = \ddot{\mathbf{v}}$$

donde $\ddot{\mathbf{v}} = \dfrac{d^2\mathbf{v}}{dt^2}$ y donde la componente j-ésima del vector $\nabla^2\mathbf{v}$ es $\left[\nabla^2 \mathbf{v}\right]_j = \dfrac{\partial^2 v_j}{\partial x_i \partial x_i}$.

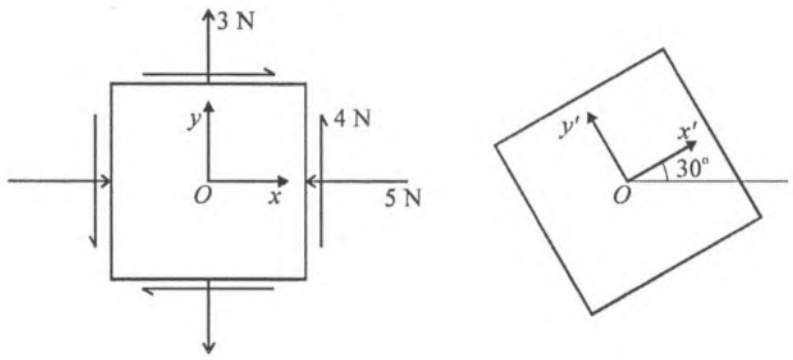
2. Sea O-$x_1x_2x_3$ un sistema de coordenadas Cartesianas rectangulares, cuya base es $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$. Se define un nuevo sistema de coordenadas Cartesianas rectangulares O'-$x_1'x_2'x_3'$ de base $\{\boldsymbol{e}_1', \boldsymbol{e}_2', \boldsymbol{e}_3'\}$, con $O'(4,-1,-2)$, $\boldsymbol{e}_1' = (2\boldsymbol{e}_1 + 2\boldsymbol{e}_2 + \boldsymbol{e}_3)/3$ y $\boldsymbol{e}_2' = (\boldsymbol{e}_1 - \boldsymbol{e}_2)/\sqrt{2}$.
   a. Exprese $\boldsymbol{e}_3'$ en términos de $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$.
   b. Dado $\boldsymbol{t} = 10\boldsymbol{e}_1 + 10\boldsymbol{e}_2 - 20\boldsymbol{e}_3$, exprese $\boldsymbol{t}$ en la base $\{\boldsymbol{e}_1', \boldsymbol{e}_2', \boldsymbol{e}_3'\}$.
   c. Exprese las coordenadas originales x_1, x_2, x_3 en términos de las nuevas x_1', x_2', x_3'.

3. Respecto a los ejes coordenados $\boldsymbol{e}_1$, $\boldsymbol{e}_2$ y $\boldsymbol{e}_1$, el tensor de tensión en un punto tiene los componentes:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

   a. Determine la componente normal σ_n y la tangencial τ del vector tracción $\overset{\boldsymbol{\nu}}{\boldsymbol{T}}$ en el plano de normal $\boldsymbol{\nu} = (\boldsymbol{e}_1 + 3\boldsymbol{e}_2 - \boldsymbol{e}_3)/\sqrt{11}$.
   b. Determine las componentes del vector unitario $\boldsymbol{e}_\tau$ tal que $\overset{\boldsymbol{\nu}}{\boldsymbol{T}} = \sigma_n \boldsymbol{\nu} + \tau \boldsymbol{e}_\tau$.

4. Sea el estado plano de tensiones de la figura de la izquierda:



   a. Calcular las tensiones y direcciones principales.
   b. Calcular dirección y magnitud de la máxima tensión de corte.
   c. Calcular el estado de tensiones correspondiente a la figura de la derecha.

1)

$$\underline{\nabla} \cdot \underline{v} = 0 \qquad \dot{\underline{v}} = \frac{dv}{dt}$$

$$\underline{\nabla} \times \underline{v} = \dot{\underline{w}} \qquad \dot{\underline{w}} = \frac{dw}{dt}$$

$$\underline{\nabla} \times \underline{w} = -\dot{\underline{v}}$$

$$\underline{\nabla^2 \underline{v} = \ddot{\underline{v}}} \; ?$$

$$\ddot{v}_i = \frac{d}{dt}\frac{dv_i}{dt} = \frac{d}{dt}\left(-\varepsilon_{ijk}\, w_{k,j}\right) =$$

$$= -\varepsilon_{ijk}\frac{\partial}{\partial x_j}\frac{d}{dt} w_k = -\varepsilon_{ijk}\frac{\partial}{\partial x_j}\varepsilon_{klm}\frac{\partial v_m}{\partial x_l} =$$

$$= -\varepsilon_{kij}\varepsilon_{klm}\frac{\partial^2}{\partial x_l \partial x_j} v_m =$$

$$= -\left(\delta_{il}\delta_{jm} - \delta_{jl}\delta_{im}\right)\frac{\partial^2}{\partial x_l \partial x_j} v_m =$$

$$= -\frac{\partial^2}{\partial x_i \partial x_m} v_m + \frac{\partial^2}{\partial x_j \partial x_j} v_i =$$

$$= -\frac{\partial}{\partial x_i}\underbrace{\frac{\partial v_m}{\partial x_m}}_{0} + \frac{\partial^2}{\partial x_j \partial x_j} v_i = \left(\nabla^2 \underline{v}\right)_i$$

2] $O - x_1 - x_2 - x_3$

$\{\underline{e}_1, \underline{e}_2, \underline{e}_3\}$

$O' - x_1' - x_2' - x_3'$ $\quad \{\underline{e}_1', \underline{e}_2', \underline{e}_3'\}$

$O'(4, -1, -2)$

$\underline{e}_1' = (2\underline{e}_1 + 2\underline{e}_2 + \underline{e}_3)/3$

$\underline{e}_2' = (\underline{e}_1 - \underline{e}_2)/\sqrt{2}$

a) $\underline{e}_3' = ?$

$\underline{e}_3' = \underline{e}_1' \times \underline{e}_2' =$

$= (2\underline{e}_1 + 2\underline{e}_2 + \underline{e}_3) \times (\underline{e}_1 - \underline{e}_2)/3\sqrt{2} =$

$= \frac{1}{3\sqrt{2}}(-2\underline{e}_1 \times \underline{e}_2 + 2\underline{e}_2 \times \underline{e}_1 + \underline{e}_3 \times \underline{e}_1 - \underline{e}_3 \times \underline{e}_2)$

$= \frac{1}{3\sqrt{2}}(-2\underline{e}_3 - 2\underline{e}_3 + \underline{e}_2 + \underline{e}_1) =$

$\underline{e}_3' = \frac{1}{3\sqrt{2}}(\underline{e}_1 + \underline{e}_2 - 4\underline{e}_3)$

b) $\underline{t} = 50\,\underline{e}_1 + 50\,\underline{e}_2 - 20\,\underline{e}_3$

$$[\underline{e}_1' \ \underline{e}_2' \ \underline{e}_3'] = [\underline{e}_1 \ \underline{e}_2 \ \underline{e}_3] \times \underbrace{\begin{bmatrix} 2/3 & 1/\sqrt{2} & 1/3\sqrt{2} \\ 2/3 & -1/\sqrt{2} & 1/3\sqrt{2} \\ 1/3 & 0 & -4/3\sqrt{2} \end{bmatrix}}_{\underline{\underline{A}}}$$

$$\underline{\underline{A}}^{-1} = \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ 1/3\sqrt{2} & 1/3\sqrt{2} & -4/3\sqrt{2} \end{bmatrix} = A^T$$

(A ortogonal)

$$\left[\underline{e}_1\ \underline{e}_2\ \underline{e}_3\right] = \left[\underline{e}'_1\ \underline{e}'_2\ \underline{e}'_3\right]\begin{bmatrix} 2/3 & 2/3 & 1/3 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ 1/3\sqrt{2} & 1/3\sqrt{2} & -4/3\sqrt{2} \end{bmatrix}$$

$$\underline{t} = \left[\underline{e}_1\ \underline{e}_2\ \underline{e}_3\right]\begin{Bmatrix} 10 \\ 10 \\ -20 \end{Bmatrix} = \left[\underline{e}'_1\ \underline{e}'_2\ \underline{e}'_3\right]\begin{bmatrix} 2/3 & 2/3 & 1/3 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ 1/3\sqrt{2} & 1/3\sqrt{2} & -4/3\sqrt{2} \end{bmatrix}\begin{Bmatrix} 10 \\ 10 \\ -20 \end{Bmatrix}$$

$$\underline{t} = \left[\underline{e}'_1\ \underline{e}'_2\ \underline{e}'_3\right]\begin{Bmatrix} 20/3 \\ 0 \\ +100/3\sqrt{2} \end{Bmatrix}$$

c)

$$\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{bmatrix} 2/3 & 1/\sqrt{2} & 1/3\sqrt{2} \\ 2/3 & -1/\sqrt{2} & 1/3\sqrt{2} \\ 1/3 & 0 & -4/3\sqrt{2} \end{bmatrix}\begin{Bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{Bmatrix} - \begin{Bmatrix} 4 \\ -1 \\ -2 \end{Bmatrix}$$

3]

$$\underline{\underline{\sigma}} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

a] σ_n y τ en plano de versor $\underline{\nu} = (\underline{e}_1 + 3\underline{e}_2 - \underline{e}_3)/\sqrt{11}$

$$\underline{T} = \underline{\underline{\sigma}}\,\underline{\nu} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} \frac{1}{\sqrt{11}} = \frac{1}{\sqrt{11}} \begin{pmatrix} 2 \\ 9 \\ -4 \end{pmatrix}$$

$$\sigma_n = \underline{T} \cdot \underline{\nu} = \frac{1}{\sqrt{11}} \begin{pmatrix} 2 \\ 9 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} \frac{1}{\sqrt{11}} = \frac{1}{11}(2+27+4) =$$

$$\sigma_n = 3$$

$$\underline{\tau} = \underline{T} - \sigma_n \underline{\nu} = \frac{1}{\sqrt{11}} \begin{pmatrix} 2 \\ 9 \\ -4 \end{pmatrix} - \frac{3}{\sqrt{11}} \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} =$$

$$= \frac{1}{\sqrt{11}} \begin{pmatrix} 2-3 \\ 9-9 \\ -4+3 \end{pmatrix} = \frac{1}{\sqrt{11}} \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

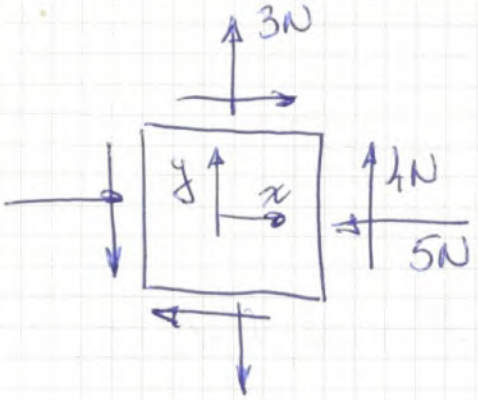
$$\tau = |\underline{\tau}| = \sqrt{\frac{2}{11}}$$

b]

$$\underline{e}_\tau = \frac{\underline{\tau}}{\tau} = \frac{1}{\sqrt{11}} \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} \frac{\sqrt{11}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

4]

a]



$$\underline{\underline{\sigma}} = \begin{bmatrix} -5 & 4 \\ 4 & 3 \end{bmatrix}$$

$$\left(\underline{\underline{\sigma}} - \sigma \underline{\underline{I}}\right) \underline{v} = 0$$

$$\det \begin{pmatrix} -5-\sigma & 4 \\ 4 & 3-\sigma \end{pmatrix} = 0$$

$$(-5-\sigma)(3-\sigma) - 16 = 0$$

$$-15 + 5\sigma - 3\sigma + \sigma^2 - 16 = 0$$

$$\sigma^2 + 2\sigma - 31 = 0$$

$$\sigma_{1,2} = \frac{-2 \pm \sqrt{4 + 4 \times 31}}{2} = \frac{-2 \pm 8\sqrt{2}}{2} = -1 \pm 4\sqrt{2}$$

$$\sigma_1 = \sigma_{max} = 4\sqrt{2} - 1 \simeq 4{,}656\ldots$$

$$\sigma_2 = \sigma_{min} = -4\sqrt{2} - 1 \simeq -6{,}656\ldots$$

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